ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 10

# MATHEMATICAL LITERACY P1

## MARKING GUIDELINE

**MARKS: 100**

**Marking guideline for the Academic Central Grade 10 Mathematical Literacy Paper 1 practice examination.**

Written by Academic Central for practice, following the CAPS curriculum for Grade 10. It is not a Department of Basic Education marking guideline.

**How to mark:** each ✓ is one mark (M = method, A = accuracy). Give follow-through marks (CA) where a learner correctly uses an earlier wrong answer. Penalise once for missing units or the R sign.

## QUESTION 1

**1.1** $2\,500 + 850 + 1\,400 + 299 + 500 + 450 + 380$ ✓ $= \text{R}6\,379$ ✓ (2)

**1.2** $6\,800 - 6\,379$ ✓ $= \text{R}421$ ✓ (2)

**1.3** $\frac{2\,500}{6\,800} \times 100$ ✓✓ $= 36{,}76\%$ ✓ (3)

**1.4** $2\,500 \times 8\% = \text{R}200$ ✓; $2\,500 + 200$ ✓ $= \text{R}2\,700$ ✓ (or $2\,500 \times 1{,}08$) (3)

**1.5** $421 - 200 = \text{R}221$ ✓; yes, she still has R221 left ✓ (2)

**1.6** $158 \times \text{R}2{,}85$ ✓ $= \text{R}450{,}30$ ✓ (2)

**1.7** Price without VAT $= 345 \div 1{,}15$ ✓ $= \text{R}300$ ✓; VAT $= 345 - 300 = \text{R}45$ ✓ (3)

**1.8.1** $149 \times 24$ ✓ $= \text{R}3\,576$ ✓ (2)

**1.8.2** $80 \times 24 = \text{R}1\,920$ ✓; $1\,999 + 1\,920$ ✓ $= \text{R}3\,919$ ✓ (3)

**1.8.3** Option A ✓, by $3\,919 - 3\,576 = \text{R}343$ ✓ (2)

**1.9** Any ONE sensible reason, e.g. for emergencies such as illness or job loss ✓✓ / to buy something big later without debt / to earn interest. (2)

**[26]**

## QUESTION 2

**2.1** 2.1.1: 2 500 m ✓ 2.1.2: 0,75 $\ell$ ✓ 2.1.3: $(3 \times 60) + 45 = 225$ min ✓ 2.1.4: 1,45 kg ✓ (4)

**2.2.1** $8 \times 6{,}5$ ✓ $= 52\ \text{m}^2$ ✓ (2)

**2.2.2** Area of one tile $= 0{,}5 \times 0{,}5 = 0{,}25\ \text{m}^2$ ✓; $52 \div 0{,}25$ ✓ $= 208$ tiles ✓ (3)

**2.2.3** $208 \div 10 = 20{,}8$ ✓ $\rightarrow 21$ boxes (round up) ✓; $21 \times \text{R}189$ ✓ $= \text{R}3\,969$ ✓ (4)

**2.2.4** $2 \times (8 + 6{,}5)$ ✓ $= 29$ m ✓ (2)

**2.2.5** $29 \times \text{R}35$ ✓ $= \text{R}1\,015$ ✓ (2)

**2.3.1** $3{,}142 \times 0{,}7^2 \times 1{,}5$ ✓✓ $= 2{,}31\ \text{m}^3$ ✓ (3)

**2.3.2** $2{,}30937 \times 1\,000$ ✓ $= 2\,309{,}37\ \ell$ (accept $2\,310\ \ell$) ✓ (2)

**2.4** $06{:}45 + 2\text{ h} = 08{:}45$ ✓; $+ 50\text{ min} = 09{:}35$ ✓ (2)

**2.5** $(1{,}8 \times 25) + 32$ ✓ $= 77\ °\text{F}$ ✓ (2)

**[26]**

## QUESTION 3

**3.1** 1 unit (cm) on the map ✓ represents 50 000 of the same units (cm) in reality ✓. (2)

**3.2** $4{,}2 \times 50\,000 = 210\,000$ cm ✓; $\div\ 100\,000$ ✓ $= 2{,}1$ km ✓ (3)

**3.3** $2{,}1 \div 5 = 0{,}42$ h ✓; $0{,}42 \times 60$ ✓ $= 25{,}2$ min (about 25 minutes) ✓ (3)

**3.4** D4 ✓ (1)

**3.5** The shop ✓ (1)

**3.6** South-east ✓✓ (2)

**3.7** $3{,}5$ cm $\times\ 100 = 350$ cm $= 3{,}5$ m ✓; $4$ cm $\times\ 100 = 4$ m ✓; area $= 3{,}5 \times 4$ ✓ $=$ $14$ m$^2$ ✓ (4)

**3.8** Yes ✓. 1 : 50 000 is a larger scale: it shows a smaller area, so more detail can be shown ✓. (2)

**3.9** So that real distances can be worked out from the map ✓✓ (accept: to show how much the area was made smaller). (2)

**[20]**

## QUESTION 4

**4.1** 19 ; 23 ; 27 ; 28 ; 30 ; 35 ; 35 ; 35 ; 41 ; 44 ; 47 ✓✓ (2)

**4.2** 35 ✓ (1)

**4.3** The 6th value (middle of 11) ✓ $= 35$ ✓ (2)

**4.4** Total $= 364$ ✓; $364 \div 11$ ✓ $= 33{,}09$ ✓ (3)

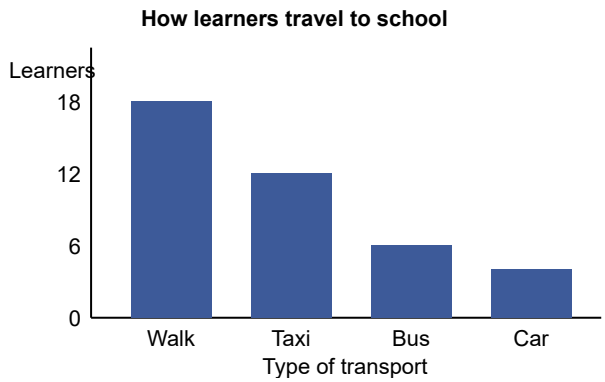
**4.5** $47 - 19$ ✓ $= 28$ ✓ (2)

**4.6** 50% of $50 = 25$ marks ✓; learners below 25: 19 and 23, so 2 failed ✓ (5)

9 passed ✓; $\frac{9}{11} \times 100$ ✓ $= 81{,}82\%$ ✓

**4.7** Median ✓. It is not pulled down by the low mark of 19 (outlier); it is the middle mark ✓. (Accept "mean" with a sensible reason, e.g. it uses all the marks.) (2)

**4.8.1** (5)




Marks: heading ✓; both axes labelled ✓; even scale ✓; correct bar heights (all four) ✓✓

**4.8.2** $\frac{18}{40} \times 100 \checkmark = 45\% \checkmark$ (2)

**4.8.3** $\frac{12}{40} \checkmark = 0{,}3 \checkmark$ (2)

**4.9** Discrete data can only take certain values (counted), e.g. number of learners ✓. (2)
Continuous data can take any value in a range (measured), e.g. height or mass ✓.

**[28]**

**TOTAL: 100**

**END OF MARKING GUIDELINE**